

TOP: Total Occupancy Guided Prediction of Binary Spectrum Tenancy

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Abstract—The applications of spectrum prediction span over a wide range of crucial fields in wireless networks, such as spectrum efficiency improvement, service quality enhancement, and network management. Despite such broad ranges of fundamental applications of spectrum prediction, existing methods are based on coarse measurement of power spectral density values. Few predictions target the actual binary tenancy of whether the spectrum slices are occupied or left unused, but their data resolution and prediction accuracy are far from satisfactory. To improve the accuracy of spectrum prediction, we propose the framework of Total Occupancy guided Prediction (TOP). It is a general prediction scheme that is flexible to incorporate an arbitrary algorithm into its framework with enhanced accuracy. Through characterizing the prediction of binary spectrum tenancy as data transmissions over the Binary Symmetric Channel (BSC), we analytically and numerically show that the two key assumptions justifying the superior performance of TOP are valid. To evaluate the accuracy of the TOP framework on spectrum tenancy from real world measurement, we set up a Software Defined Radio (SDR) testbed to measure LTE spectrum tenancy by decoding the Downlink Control Information (DCI) to gain high resolution usage at the same granularity with LTE scheduling. Armed with the high resolution data, we adapt the Multi-Layer Perceptron (MLP) algorithm into the TOP framework to validate its performance. The thorough experiments reveal that the TOP framework significantly improves MLP accuracy from 0.84 to 0.91, outperforming many state-of-the-art prediction schemes.

Index Terms—spectrum tenancy prediction, dynamic spectrum access, data granularity.

I. INTRODUCTION

Radio spectrum has been a fundamental resource for wireless communications, and the prediction of its usage is becoming more and more important for spectrum efficiency, service qualities, and network management [1]. One important scheme to enhance spectrum efficiency is to allow the opportunistic exploitation of unoccupied spectrum slices in the bands that used to be exclusively assigned. To avoid collisions in such Dynamic Spectrum Access (DSA) systems [2], cognizant transceivers can hugely benefit from accurate spectrum usage predictions. In addition to improving spectrum efficiency, spectrum tenancy prediction has many desirable applications. For example, it can enhance the service quality of mobile video streaming through adaptive encoding based on inferred available bandwidth in LTE cells [3]. Additionally,

spectrum tenancy prediction can assist mobile operators in designing and managing reliable cellular networks [4]–[7].

Due to the broad impacts of spectrum tenancy prediction, extensive research has been conducted in this area, covering predictions of spectrum usage in terrestrial wireless networks as well as space information networks [8]–[10]. Despite various investigations into spectrum tenancy inference, current predictions of spectrum usage have two major drawbacks. The first issue is that existing prediction algorithms are based on Power Spectral Densities (PSD) with coarse time-frequency granularities, making it uncertain whether they can be used to predict binary tenancy (i.e., whether a slice of spectrum resource is occupied or unused) in the actual scheduling resolutions. For instance, the prediction algorithm in [11] targets the spectrum tenancy in GSM bands, but the training data has a time resolution of 180 seconds, during which the spectrum usage of a GSM system may change hundreds of times with GSM bursts as short as 0.577 ms. While a few studies have explored prediction of binary tenancy instead of continuous PSD values, their prediction accuracy is unsatisfactory. For example, three state-of-the-art prediction algorithms for binary spectrum tenancy achieve accuracy in the range of 54.82% to 65.09% [12]. Such accuracy is not adequate for binary predictions, as random guessing would achieve 0.5 accuracy for spectrum tenancy with equal amounts of idle and busy slices.

To address the aforementioned issues in spectrum tenancy prediction, we answer the research question of how to accurately predict binary spectrum tenancy in high time-frequency resolutions in this paper. We first adopt the decoding based method to investigate the binary spectrum tenancy of an LTE cell in its scheduling unit, i.e., 1 ms by 180 kHz [13]. Different from the widely adopted energy detection method that measures the PSD values that requires an artificial choice of a threshold value to decide binary tenancy, the tenancy status assigned by the base stations can be decoded this way, and the binary data set will serve the training and the validation purposes. After gathering the binary spectrum tenancy in fine time-frequency granularity, we propose the Total Occupancy guided Prediction (TOP) framework. Specifically, TOP decides either to predict the spectrum with an algorithm of choice, or to adapt the algorithm into the Total Regulated Prediction (TRP). There are two major steps in

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the TRP algorithm. First, the total number of occupied LTE Resource Blocks (RB) in a scheduling interval is predicted. Next, we predict the tenancy of each LTE RB individually with an arbitrary algorithm to obtain their probabilities of being occupied. Then, the RBs are ranked based on their probabilities of being busy from high to low, and the ones ranked before the predicted total number of busy RBs are deemed occupied. In this way, the accuracy of the prediction algorithm is improved. For example, the accuracy of directly predicting the channel occupancy is improved by the TOP framework in Section IV from 0.84 to 0.91.

In order to gather the high resolution binary spectrum tenancy, we deploy an LTE spectrum measurement system as proposed in [13]. The binary tenancy is parsed from the LTE Downlink Control Information (DCI) decoded by the Software Defined Radio (SDR) system. Since the usage of LTE spectrum resources in a cell is centrally assigned by the base stations and then announced in the DCIs, the binary tenancy is achieved at the same time-frequency granularity as LTE scheduling through DCI decoding. Armed with the spectrum tenancy in its true granularity, we are able to train and validate the proposed TOP prediction framework with high resolution data in the same granularity as scheduling, leading to more applicable predictions than other algorithms based on coarse data [11].

After obtaining the high resolution tenancy, we propose the TOP framework and its design rationale. The analytical studies on TOP and the numerical results prove that the framework improves the prediction accuracy under reasonable assumptions, regardless of the underlying prediction algorithms for each single RB and the total number of occupied RBs. This justifies the wide range of potential applications of the TOP framework. Then, we adopt the Multi-Layer Perceptron (MLP) algorithm to predict the tenancy of each LTE RB and the total number of occupied RBs. The prediction accuracy of MLP alone and that of employing MLP in the TOP framework is compared in the performance evaluations, where the TOP framework boosts the accuracy by 8%.

We recapitulate the contributions of this paper as follows.

- To investigate the prediction algorithms for spectrum tenancy in high time-frequency resolutions, we set up an SDR test bed to gather LTE spectrum tenancy in the same granularity as scheduling. The high resolution tenancy serves as the training and validation data.
- We propose a heuristic prediction framework TOP that is compatible with various prediction algorithms to enhance the overall prediction accuracy. The design rationale of TOP is investigated both analytically and numerically.
- In performance evaluations, we adapt MLPs into the TOP framework to prove that TOP is able to improve the prediction accuracy of state-of-the-art prediction algorithms.

The remainder of this paper is organized as follows. In Section II, the TOP framework is described and analyzed. We evaluate the performance of TOP in Section IV. Then, we introduce how we gather the high resolution spectrum tenancy in Section III. The paper is concluded in Section V.

II. THE TOP PREDICTION FRAMEWORK

In this section, we introduce the TOP procedures, and analytically explain the design rationale behind the TOP framework that improves prediction accuracy of multiple RBs.

A. The TOP framework

The TOP framework is versatile and effective in that it is able to adapt various spectrum predictions into the framework with improved accuracy. As a prediction framework, TOP employs other algorithms to predict spectrum tenancy and the total number of occupied RBs. Then, the TRP algorithm is proposed to refine the binary predictions for each RB by the inferred total number of occupied RBs. The TOP framework then decides whether the directly predicted tenancy or the TRP regulated forecast will be adopted as the results.

The prediction input $\mathbf{A}$ is a matrix of previous spectrum tenancy. The prediction target is a vector of spectrum usage of all the LTE RBs in a scheduling interval, and they are predicted as $\mathbf{T} = \langle T_1, \dots, T_n \rangle$, where n is the number of RBs. A prediction algorithm that infers the binary tenancy based on the given input can be denoted as $\mathcal{P}_b(\mathbf{A}) = \langle p_1, \dots, p_n \rangle$, where the predicted probabilities of being busy are converted to the binary tenancy vector $\mathbf{T}$ by choosing a threshold value h . Besides the tenancy prediction algorithms $\mathcal{P}_b$, the TOP framework requires the prediction of the total number of occupied RBs to guide the tenancy prediction. Denote the prediction of the total number of occupied RBs as $\mathcal{P}_a(\mathbf{A}) = S$, where $S = \sum_{i=1}^n T_i$ after running the TRP algorithm as introduced below.

Algorithm 1 The TRP algorithm

Input: Prediction algorithms $\mathcal{P}_a, \mathcal{P}_b$, and their input $\mathbf{A}$.

Output: The predicted tenancy $\mathbf{T} = \langle T_1, \dots, T_n \rangle$.

- 1: Apply algorithm $\mathcal{P}_b$ to obtain the probabilities of being occupied, $\langle p_1, \dots, p_n \rangle$ of all RBs.
 - 2: Employ algorithm $\mathcal{P}_a$ to acquire the predicted total number of busy RBs S .
 - 3: **for** $i = 1, 2, \dots, n$ **do**
 - 4: **if** p_i ranks among the top S in $\langle p_1, \dots, p_n \rangle$ **then**
 - 5: Mark the i th RB as occupied, and set $T_i = 1$.
 - 6: **else**
 - 7: Mark the i th RB as idle by setting $T_i = 0$.
 - 8: **end if**
 - 9: **end for**
-

1) *The TRP algorithm:* Instead of converting the probabilities of RBs being occupied delivered by $\mathcal{P}_b$ according to a threshold value, the TRP algorithm relies on the inferred total number of occupied RBs produced by $\mathcal{P}_a = S$ to convert tenancy probabilities to the binary results of either being occupied or not. Then, the RBs with the largest S probabilities in $\mathcal{P}_b(\mathbf{A}) = \langle p_1, \dots, p_n \rangle$ of being occupied are considered as busy. The TRP algorithm is also described in Algorithm 1. Compared with an artificial choice of the threshold value h to convert the occupancy probabilities obtained by $\mathcal{P}_b$ to

binary, the TRP algorithm decides the tenancy based on the more intuitive number of total busy RBs predicted by $\mathcal{P}_a$.

2) *The TOP framework*: Compared with predicting the tenancy with $\mathcal{P}_b$ and converting to binary based on a threshold, TRP may introduce additional errors if the prediction of the total tenancy is of low accuracy. Thus, we propose the TOP framework to use $\mathcal{P}_b$ when TRP does not predict more accurately, and TRP is only enabled if it outperforms the classical way of adopting $\mathcal{P}_b$ and a conversion threshold.

To realize the aforementioned TOP framework, another binary classifier $\mathcal{C}$ needs to be trained to decide whether the TRP or the predictor $\mathcal{P}_b$ is more promising for a given prediction input $\mathbf{A}$. The prediction inputs for the classifier are the same binary spectrum tenancy as the prediction features for $\mathcal{P}_b$. The training targets of $\mathcal{C}$ are 0 when $\mathcal{P}_b$ outperforms the TRP, and 1 while the TRP prevails.

After $\mathcal{C}$ is trained, the TOP framework predicts the tenancy as explained in Algorithm 2. First, the TOP decides which algorithm to use with the classifier $\mathcal{C}$. Then, the binary spectrum tenancy is inferred by the chosen algorithm.

Algorithm 2 The TOP framework

Input: Prediction algorithms $\mathcal{P}_a, \mathcal{P}_b, \mathcal{C}$, their input $\mathbf{A}$, and the TRP algorithm.

Output: The predicted tenancy $\mathbf{T} = \langle T_1, \dots, T_n \rangle$.

- 1: Apply algorithm $\mathcal{C}$ to decide which algorithm to use.
 - 2: **if** $\mathcal{C}(\mathbf{A}) == 0$ **then**
 - 3: Predict $\mathbf{T}$ with the algorithm $\mathcal{P}_b$.
 - 4: **else**
 - 5: Call Algorithm 1 to predict $\mathbf{T}$ with inputs $\mathcal{P}_a, \mathcal{P}_b, \mathbf{A}$.
 - 6: **end if**
-

B. The TOP design rationale

In this subsection, we reveal the design rationale behind the TOP framework, and answer the question under what conditions it predicts more accurately than directly using the algorithm $\mathcal{P}_b$. After examining the TOP framework, we discover that the validity of TOP is based on two important assumptions. The first one is that the classification algorithm $\mathcal{C}$ is able to effectively identify the more accurate predictor between the TRP and the $\mathcal{P}_b$ algorithms for sample $\mathbf{A}$. The second assumption is that the TRP algorithm predicts more accurately than the $\mathcal{P}_b$ for some input $\mathbf{A}$.

1) *Analysis of assumption 1*: To examine the first assumption, we model the classification $\mathcal{C}$, and the tenancy predictions of both $\mathcal{P}_b$ and TRP as the Binary Symmetric Channel (BSC). We denote the probability that the classification $\mathcal{C}$ chooses the right algorithm by p_c . For tenancy $T_i \in \mathbf{T}$, the probability of correct prediction by $\mathcal{P}_b$ is $p_{b,l}$ when T_i is in the group of samples with probability (WP) of l_b where $\mathcal{P}_b$ has low prediction accuracy. The accuracy is $p_{b,h}$ WP $1 - l_b$ when T_i belongs to another group where $\mathcal{P}_b$ has higher accuracy. Similarly, we consider that the TRP has two different prediction accuracy of $p_{T,l}$ and $p_{T,h}$, for l_T and $1 - l_T$ of the total number of samples, respectively. Hence,

$p_{b,l} < p_{b,h}$ and $p_{T,l} < p_{T,h}$. For brevity, we further assume that $p_{T,l} < p_{b,h} < p_{T,h}$ in the following analysis, and the results can be derived similarly in other cases. Under these assumptions, the probability of the correct prediction of T_i by algorithm $\mathcal{P}_b$ is

$$\alpha = l_b p_{b,l} + (1 - l_b) p_{b,h}, \quad (1)$$

denoted as α . The probability of correct prediction by the TOP framework is

$$\begin{aligned} \beta = & l_b(1 - p_c)p_{b,l} + l_b p_c(l_T p_{T,l} + (1 - l_T)p_{T,h}) \\ & + (1 - l_b)l_T p_c p_{b,h} + (1 - l_b)l_T(1 - p_c)p_{T,l} \\ & + (1 - l_T)p_c p_{T,h} \\ & + (1 - l_T)(1 - p_c)(l_b p_{b,l} + (1 - l_b)p_{b,h}), \end{aligned} \quad (2)$$

denoted as β . By comparing the two probabilities of α and β , we find the condition that the TOP framework outperforms simply inferring with algorithm $\mathcal{P}_b$ is that $p_c \geq \gamma_n/\gamma_d$, where the constant γ_n, γ_d is as follows.

$$\begin{aligned} \gamma_n = & -l_b p_{b,l} - l_T p_{T,l} + l_T p_{b,h} \\ & + l_b l_T p_{T,l} + l_b l_T p_{l,b} - l_b l_T p_{b,h}, \\ \gamma_d = & p_{T,h} - p_{b,h} - l_T p_{T,h} + 2l_T p_{b,h} \\ & + l_b p_{T,h} - 2l_b p_{b,l} - l_T p_{T,l} + l_b p_{b,h} \\ & + 2l_b l_T p_{T,l} - l_b l_T p_{T,h} - 2l_b l_T p_{b,h} + l_b l_T p_{b,l} \end{aligned} \quad (3)$$

Remarks: For given prediction algorithms $\mathcal{P}_b$ and TRP, there is always a classification algorithm $\mathcal{C}$ that is able to identify the better algorithm to use and achieve the improved accuracy in the TOP algorithm. To achieve this goal, the classifier $\mathcal{C}$ needs to achieve accuracy of at least γ_n/γ_d . Thus, the first assumption of the TOP framework is validated.

2) *Analysis of assumption 2*: The other important assumption of the TOP framework is that the TRP algorithm predicts more accurately than $\mathcal{P}_b$, at least sometimes. To validate this assumption, we denote the probability of tenancy T_i to be 0 as a_t for all $i \in \{1, \dots, n\}$, and $1 - a_t$ otherwise. Let the accuracy of the TSC model for the prediction of T_i as p_i , and we assume that $p_1 \geq p_2 \geq \dots \geq p_n$. Denote the predicted tenancy by algorithm $\mathcal{P}_b$ as random variables $X_i, i \in \{1, \dots, n\}$. Thus, the prediction accuracy of $\mathcal{P}_b$, denoted as π , can be written as $\pi = \mathbb{E}(\sum_{i=1}^n \mathbb{1}_{\{X_i=T_i\}})/n$, where $\mathbb{E}()$ is the expectation operator and $\mathbb{1}_{\{\cdot\}}$ is the indicator function, so $\pi = \sum_{i=1}^n p_i/n$.

Let the prediction of the TRP algorithm be the random variable Y_i , so the accuracy of the TRP is defined as $\theta = \mathbb{E}(\sum_{i=1}^n \mathbb{1}_{\{Y_i=T_i\}})/n$. Since the key is to prove that TRP is possible to achieve higher prediction accuracy than $\mathcal{P}_b$ in certain cases, we consider the special case when $\mathcal{P}_a$ is always correct, i.e. $\sum_i Y_i = \sum_i T_i$ for tractability. In this case, the TRP accuracy can be expressed as

$$\theta = \frac{1}{n} \sum_{c=0}^n \sum_{d=0}^n \mathbb{E} \left(\sum_i \mathbb{1}_{\{Y_i=T_i\}} \mid \sum X_i = c, \sum T_i = d \right). \quad (4)$$

Based on the above equation, θ can be expressed in terms of p_i , and numerically computed by adding up the expectations when c, d take different values.

To demonstrate that the TRP is able to outperform $\mathcal{P}_b$ in certain situations, we provide the following numerical study. In this specific case, we consider that there are five RBs in total, i.e. $n = 5$, and prediction accuracy of $\mathcal{P}_b$ on the five RBs are $p_i = 1 - 0.1i, i \in \{1, \dots, n\}$. Thus, $\pi = 0.7$, regardless of other parameters. As θ is further dependent on the probability of being occupied, we consider different values of a_t ranging from 0.1 to 0.9 at 0.1 intervals. Then, we obtain the numerical comparisons of the prediction accuracy between the TRP and that of directly using $\mathcal{P}_b$, shown in Fig. 1. We observe that TRP outperforms $\mathcal{P}_b$ for all tenancy probabilities, and the advantages are more pronounced when the tenancy becomes more unbalanced. The reason is that TRP has a better chance to flip the wrong prediction of $\mathcal{P}_b$ when the tenancy is naturally more inclined towards idle or busy. Even if the tenancy is strictly neutral with exactly $a_t = 0.5$, TRP still obviously outperforms with the accuracy of 0.7839 over 0.7.

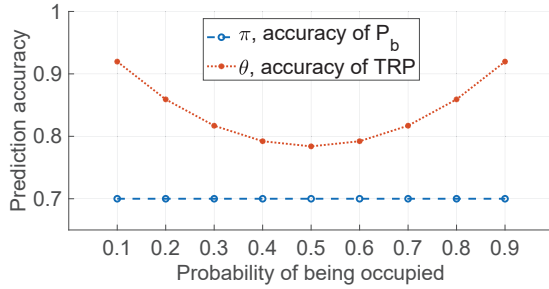


Fig. 1. A numerical study on accuracy comparisons between $\mathcal{P}_b$ and TRP.

Remarks: We have proved that the TRP algorithm is able to predict spectrum tenancy more accurately than $\mathcal{P}_b$ in some cases based on the analytical and numerical studies. Thus, the second assumption of the TOP framework is validated.

C. Complexity of the TOP framework

As previously introduced, three prediction algorithms are involved in the TOP framework, the algorithm $\mathcal{P}_b$ that predicts the binary tenancy of the n RBs, the algorithm $\mathcal{P}_a$ that predicts the total number of occupied RBs, and the classifier $\mathcal{C}$ that predicts whether $\mathcal{P}_b$ or TRP should be used to predict the tenancy of multiple RBs in the next time slot. Without the TOP framework, the complexity of the prediction algorithm is only dependent on $\mathcal{P}_b$. Because $\mathcal{P}_b$ predicts the tenancy of n RBs that require independent prediction, we consider that the complexity of $\mathcal{P}_b$ scales linearly with the number of RBs, i.e. $O(n)$. Within the TOP framework, two additional predictions, $\mathcal{P}_a$ and $\mathcal{C}$, are conducted based on the same inputs as $\mathcal{P}_b$, to produce a single output, respectively. Hence, the TOP complexity is $O(n+2) = O(n)$, assuming that $\mathcal{P}_a$ and $\mathcal{C}$ adopt algorithms with similar complexity to the prediction for a single RB. To sum up, the complexity of the TOP framework scales linearly with the number of RBs, i.e., $O(n)$. This is

commensurate with that of other algorithms which can be adopted as $\mathcal{P}_b$ to independently forecast RB tenancy.

In this section, we have proposed the heuristic prediction framework TOP, and validated the two assumptions of TOP to prove that it is guaranteed to improve the prediction accuracy of the prediction algorithm $\mathcal{P}_b$ of general choices.

III. PREDICTION FEATURES AND DATA COLLECTION

Since the data is of vital importance for prediction algorithms [14], we explain the prediction features for LTE spectrum tenancy prediction and how we collect the spectrum tenancy data.

A. Prediction features

The prediction features or inputs to prediction algorithms of spectrum tenancy have been traditionally considered as the tenancy in previous time instances [8]. Thus, the prediction features for LTE spectrum tenancy are the historical usage of LTE RBs. In the frequency domain, the spectrum resources of an LTE cell are orthogonally split into multiple RBs with the bandwidth of 180 kHz, and the RBs are dynamically scheduled every millisecond. Thus, the prediction features are considered as the binary tenancy of LTE RBs in a specific number of previous milliseconds.

B. Data collection

Based on the decoding method proposed in [13], we set up a measurement system and gather spectrum tenancy data to obtain LTE spectrum usage at fine granularities. Because the DCI of LTE systems is not protected by any encryption schemes, the DCIs can be parsed by sniffers. LTE base stations broadcast to all the users the decisions on how the spectrum resources are assigned every millisecond in the DCIs via a physical layer control channel in clear texts. Thus, the control information can be sniffed and parsed to reveal the assignment of spectrum resources at the same time and frequency granularities as LTE scheduling.

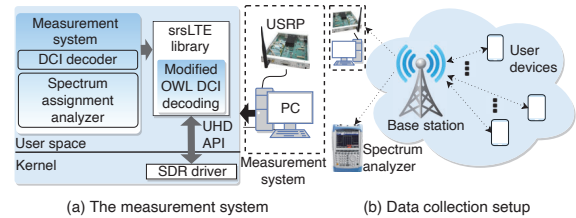


Fig. 2. The measurement system and the measurement setting.

The spectrum tenancy decoder is implemented in the user space of a Linux computer. There are two main function blocks, one for decoding the DCIs, and the other for parsing the spectrum tenancy in the DCI messages. We call the srsLTE library for existing LTE routines [15], including the DCI decoding functions provided by OWL [16]. The radio front end is realized by an SDR system. Through the SDR driver API, the decoder calls USRP Hardware Driver (UHD) version 3.9.7 to communicate with the SDR device. The SDR system

includes a USRP X310 mother board and two SBX-120 wide-band daughter-boards. The SDR boards contain function blocks to convert analog signals to complex samples. The host computer has a quad-core CPU and 16 GB memory, running Ubuntu 16.04. The software and the hardware configurations of the measurement tool are also depicted in Fig. 2(a).

Fig. 2(b) illustrates the measurement setting. A spectrum analyzer is employed to search and verify the existence of an LTE cell with the best signal to noise ratio in the vicinity of our lab. The downlink system bandwidth of the cell is 10 MHz that accommodates 50 LTE RBs which will also be referred to as channels interchangeably. These channels are frequency-wise orthogonal to one another, and each occupy 180 kHz. We collect LTE spectrum tenancy data of the cell for 24 hours, obtaining a total of 4.32×10^9 pieces of binary spectrum tenancy, each of which is either marked as unused by zero, or occupied by one.

IV. PERFORMANCE EVALUATIONS

The performance of the TOP framework is thoroughly studied and compared with different algorithms in this section.

As the TOP is a prediction framework that enhances the accuracy of prediction algorithms in general, the first step is to decide on specific algorithms for predictors in the framework. Among the wide variety of spectrum prediction algorithms [17], we choose the MLP algorithm as the predictors for $\mathcal{P}_a$, $\mathcal{P}_b$ and the classifier $\mathcal{C}$ in the TOP framework. The reasons for employing the MLP are twofold. First, the MLP prediction has achieved high accuracy when the training data are PSD values in low resolutions [18], so it is likely that MLP is able to predict accurately for high resolution binary data if the underlying logic behind the data is the same. Second, since the TOP is an accuracy improvement framework, it is more important that TOP is able to further boost the accuracy of algorithms that are able to predict accurately on themselves, than improving the accuracy of low precision predictions.

The accuracy of TOP is compared with three other algorithms. One is to directly predict with $\mathcal{P}_b$ that is a trained MLP. This scheme is named Dir, and such a comparison is going to show whether TOP improves the accuracy of a chosen $\mathcal{P}_b$. Another comparison algorithm is to employ the MLP that takes the total number of occupied RBs in a time slot as an extra prediction feature, named as WT in the following. As we will see from the results, the accuracy is not much improved by simply predicting the tenancy with the total occupancy as a new feature of the MLP in addition to binary tenancy. The third scheme to compare with is the TRP that always regulates prediction results of MLP $\mathcal{P}_b$ with the predicted total tenancy.

The metrics for performance evaluations is the average accuracy. As we train and apply the four prediction schemes into 24 groups of spectrum tenancy data collected under three different levels of wireless traffic, the accuracy is then averaged across all the cases.

The prediction features are of key significance for all algorithms, so we first investigate how many previous spectrum

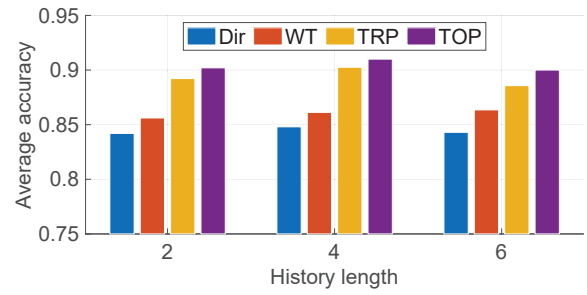


Fig. 3. The impact of the size of the history tenancy for $\mathcal{P}_b$.

usage should be considered as the prediction inputs for the four prediction schemes. Because LTE systems dynamically schedule the spectrum usage every one millisecond and our prediction is in the same time granularity, we consider small numbers of previous tenancy because the correlations fade away fast as reported in other studies [19]. As illustrated in Fig. 3, three different numbers (two, four, and six) of previous spectrum occupancy are used as prediction features for $\mathcal{P}_b$ in the four schemes. All the schemes achieve higher prediction accuracy when four previous tenancy serve as the prediction inputs. Thus, we use four history tenancy as prediction features for $\mathcal{P}_b$ for the rest of the performance evaluations. In terms of accuracy comparisons among the four schemes, the TOP framework significantly improves the accuracy of Dir and performs the best.

TABLE I
ACCURACY VS. THE NUMBER OF NEURONS IN THE HIDDEN LAYER.

Neuron count	Dir	WT	TRP	TOP
4	0.8472	0.8561	0.8922	0.9047
8	0.8409	0.8538	0.8898	0.8962
12	0.8423	0.8581	0.8890	0.9073

Since the workhorse of the four predictions is the MLP $\mathcal{P}_b$, we study the structure of the neural network. Since there are four neurons in the input layer and one in the output, we deploy one hidden layer and choose the number of neurons in the layer through trial and error. Table I shows the performance of the four schemes with the MLP $\mathcal{P}_b$ having three different numbers of neurons in the hidden layer. Because the MLP with four neurons in the hidden layer achieves the highest accuracy, such a structure is adopted due to its accuracy and the simple structure that saves computing resources during the training stage. The proposed TOP framework again outperforms all other schemes regardless of the $\mathcal{P}_b$ structures.

TABLE II
ACCURACY VS. THE NUMBER OF TRAINING SAMPLES.

Sample count	Dir	WT	TRP	TOP
50	0.8416	0.8561	0.8922	0.9047
100	0.8322	0.8388	0.8850	0.8962
300	0.8097	0.8161	0.8787	0.8873
500	0.8238	0.8168	0.8825	0.9037

Next, we look into the impacts of the number of training

samples on prediction accuracy. We train the four prediction schemes with four different numbers (50, 100, 300, 500) of samples as depicted in Table II. We observe that the four schemes reach the best prediction results when the training samples contain 50 data points. This outcome suggests that the four schemes can be trained by small data sets collected in only 0.05 seconds to obtain the best prediction accuracy. As long as the spectrum usage data can be collected in time, retraining $\mathcal{P}_b$ online is not an expensive computation task.

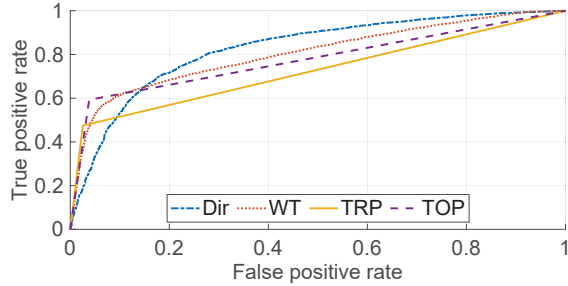


Fig. 4. ROC comparisons for the four prediction algorithms.

Besides prediction accuracy, we study the Receiver Operating Characteristic (ROC) curve in Fig. 4. ROC shows the trade-off between the false positive and the true positive rates, and the good ROC curve should be close to the point (1, 0). As shown in the figure, the proposed TOP algorithm achieves a significantly better true positive rate than only using the MLP. The ROC of TOP is the closest to the ideal ROC in comparison with the other three algorithms.

To sum up, the MLP algorithm achieves satisfactory accuracy when applied to binary spectrum tenancy prediction compared with the around 0.6 accuracy obtained by classical methods [12]. Our proposed TOP framework is able to further enhance the prediction accuracy of the MLP from around 0.84 to as high as 0.91, highlighting its effectiveness and capability to enhance a general prediction algorithm. The TOP framework outperforms all the other three schemes including the WT, meaning that simply adding the total tenancy as a new prediction feature cannot reach the accuracy of the TOP. Additionally, we discover that the TOP does not require a large neural network and can be trained with a data set as small as 50 samples.

V. CONCLUSION

We propose the general prediction framework TOP for binary spectrum tenancy. This framework is able to incorporate any prediction algorithms into its structure and improve their prediction accuracy. This statement is based on two key assumptions which have been analytically and numerically proved. The performance of the TOP framework is thoroughly evaluated based on binary spectrum tenancy measured from real world commercial LTE systems. In order to validate the TOP framework with high resolution spectrum usage in the same time-frequency granularity as LTE scheduling, we set up an SDR testbed and adopt the decoding based method for LTE spectrum measurement to achieve the binary tenancy in the

time-frequency unit of 1 ms by 180 kHz. In the performance evaluations, we show that the accuracy of TOP outperforms three other prediction schemes. TOP is able to further improve the prediction accuracy of the state-of-the-art MLP algorithm.

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